

K1–AM–SF: A Tailored Gini-Interval Branch-and-Cut Framework for Exact Static Rebalancing

Algorithm chapter generated from the audited stable implementation

August 29, 2026

Abstract

We present K1–AM–SF, the stable paper-facing exact method selected by `paper-k1-am-sf`. The tailored Gini-interval branch-and-cut framework starts from one complete improving interval, uses midpoint adaptive-mass refinement with threshold 0.08, and solves each unresolved leaf with the F0–CLEAN sparse tailored fixed-interval MILP under Gurobi’s native branch-and-cut. Dynamic user-cut callbacks and custom branching are off. Exact interval coverage, scope-valid bounds, and independent incumbent verification govern certification. A bounded inventory–route cut study strengthened root relaxations but failed the fixed-interval promotion gate and remains outside the stable algorithm.

1 Problem Definition

Let $\mathcal{I} = \{1, \dots, n\}$ be the stations, let $\mathcal{K}$ be the vehicles, and let 0 denote the depot. Station i has initial inventory b_i , capacity c_i , positive target t_i , and penalty weight ω_i . Vehicle k has capacity Q_k and duration limit T_k . Travel times are τ_{ij} .

The compact model uses binary arc variables x_{kij} and visit variables z_{ki} , nonnegative integer pickup and drop quantities p_{ki} and d_{ki} , and final station inventory

$$Y_i = b_i + \sum_{k \in \mathcal{K}} (d_{ki} - p_{ki}), \quad 0 \leq Y_i \leq c_i. \quad (1)$$

Standard depot flow, station flow, visit–service linking, station disjointness, vehicle load conservation, and load-capacity constraints make the selected arcs and operations an original feasible route-load plan. Vehicles start empty and may return residual load to the depot. Under the implemented handling convention, route duration is

$$\sum_{i,j} \tau_{ij} x_{kij} + c^{\text{unit}} \sum_i p_{ki} \leq T_k, \quad c^{\text{unit}} = c^{\text{pick}} + c^{\text{drop}}. \quad (2)$$

Thus a global handling inequality may charge pickups only; charging both p and d by c^{unit} would not be valid for this convention.

Define the target ratios, denominator, pairwise deviations, and numerator by

$$r_i = Y_i/t_i, \quad S = \sum_i r_i, \quad h_{ij} = |r_i - r_j| \ (i < j), \quad H = \sum_{i < j} h_{ij}. \quad (3)$$

Because $t_i > 0$ and the feasible model maintains positive aggregate station inventory, $S > 0$. The Gini term [1] and weighted target-deviation penalty are

$$G = \frac{H}{nS}, \quad P = \sum_i \omega_i e_i, \quad qqquad e_i = |r_i - 1|. \quad (4)$$

For fixed $\lambda \geq 0$, the objective is

$$F = G + \lambda P. \quad (5)$$

The binary-expansion compact formulation represents the products required by the Gini objective to the declared numerical tolerance. An independently reconstructed route-load plan is evaluated directly with (4)–(5); only such a verified plan can supply an upper bound.

Definition 1 (Original feasible set). $\mathcal{X}$ is the set of route, service, load, and final-inventory vectors that satisfy the original compact routing and operation constraints and the duration convention (2). Strengthening rows do not redefine $\mathcal{X}$; each paper-core row below is necessary for membership in $\mathcal{X}$ under its stated interval condition.

2 Tailored Gini-Interval Branch-and-Cut Framework

Suppose an independently verified incumbent has objective $\bar{F}$. Since $P \geq 0$, every improving solution has $G < \bar{F}$. K1–AM–SF begins with one closed interval ($K_0 = 1$) covering the complete improving Gini domain. It maintains a ledger of closed intervals $\Gamma_q = [\gamma_q^L, \gamma_q^U]$ whose union covers that domain. Unresolved leaves are refined at their exact midpoints using the adaptive-mass score and the frozen threshold $\tau = 0.08$. Native-target handling, exact-parent closure, and child-infeasibility handling are inherited unchanged from the qualified controller.

For each leaf, the sparse tailored fixed-interval MILP solved by Gurobi’s native branch-and-cut enforces

$$\gamma_q^L \leq G \leq \gamma_q^U, \quad (6)$$

$$G + \lambda P \leq \bar{F} - \varepsilon. \quad (7)$$

It is the F0–CLEAN formulation: the Round 50 v0 model with only the historical exhaustive $V \leq 12$ subset-duration block removed. A timeout or wrapper checkpoint never closes a leaf by itself.

Proposition 2 (Fixed-interval no-improver validity). *If the original compact model together with (6), (7), and valid static strengthening is infeasible, then no original feasible solution in Γ_q improves the incumbent.*

Proof. Every original improving solution in Γ_q satisfies the original model, the interval and cutoff rows, and every valid strengthening row. Such a solution would contradict infeasibility. $\square$

Theorem 3 (Global exactness). *Let $\hat{x}$ be independently verified with objective $\bar{F}$. If the final leaves exactly cover every improving Gini value, every leaf is closed by a scope-valid bound, exact fixed-interval infeasibility, valid fixed-interval best bound, or empty-domain proof, and no invalid or open leaf remains, then $\hat{x}$ is globally optimal.*

Proof. Any better solution has a Gini value in one final leaf, whose valid closure excludes that improving solution, a contradiction. $\square$

Operationally, the framework verifies a same-run incumbent, initializes the complete K1 ledger, propagates interval domains, solves unresolved leaves with F0–CLEAN, refines only through the frozen midpoint/adaptive-mass rule, merges only scope-valid evidence, and certifies only after coverage and independent solution audits pass. The stable algorithm registers no dynamic user-cut callback.

3 Active Sparse Fixed-Interval Formulation

The active families below are generated by the canonical `paper-k1-am-sf` preset. They are static rows, variable domains, or an extended formulation; none is a dynamic user-cut callback.

3.1 Interval and product structure

For $G = H/(nS)$ and $S > 0$, every fixed interval satisfies

$$H \leq n\gamma^U S, \quad H \geq n\gamma^L S. \quad (8)$$

If $a \in \{0, 1\}$, $G \in [\gamma^L, \gamma^U]$, and $w = Ga$, the four rows

$$w \geq \gamma^L a, \quad w \leq \gamma^U a, \quad (9)$$

$$w \geq G - \gamma^U(1 - a), \quad w \leq G - \gamma^L(1 - a) \quad (10)$$

give the exact binary product hull. K1-AM-SF applies this hull to the binary expansion of final inventory, together with tight denominator bounds.

Let valid domains be $P \in [P^L, P^U]$ and $S \in [S^L, S^U]$. The auxiliary W_{SP} uses the four McCormick rows for SP and the paper-safe estimator

$$H + n\lambda W_{SP} \leq n(\bar{F} - \varepsilon)S. \quad (11)$$

The simpler objective estimator and the penalty lower-bound closure are also active. Final-inventory penalty domains, movement-reachability domains, and two deterministic domain-propagation rounds tighten bounds without removing any original feasible point.

3.2 Inventory, visits, duration, and connectivity

Inventory conservation includes

$$\sum_i b_i - \sum_k Q_k \leq \sum_i Y_i \leq \sum_i b_i. \quad (12)$$

The visit links

$$Y_i - b_i \leq (c_i - b_i) \sum_k z_{ki}, \quad (13)$$

$$b_i - Y_i \leq b_i \sum_k z_{ki} \quad (14)$$

force $Y_i = b_i$ when no route visits station i . Exact small-support travel lower bounds generate pair and triple support-duration covers. A vehicle-indexed connectivity-flow extended formulation links selected arcs and visits. The verified-incumbent objective row is an upper-bound cutoff only; it does not import lower-bound evidence.

The exhaustive station-subset duration block is deliberately absent. Gini spread, required-movement, transfer-cutset, subset-inventory, dynamic support-duration, and all other tailored callback families are inactive in the stable preset. Earlier sections or experiments using those families are historical and are not evidence about K1-AM-SF.

3.3 Audited inventory–route research family

Round 54 separately considered, for nonempty station A excluding the depot,

$$\sum_{i \in A} (Y_i - b_i) \leq \sum_k Q_k \delta_k^-(A), \quad (15)$$

$$\sum_{i \in A} (b_i - Y_i) \leq \sum_k Q_k \delta_k^+(A). \quad (16)$$

Summing vehicle load conservation inside A shows that net delivery is at most the load entering A , and entering load is at most route capacity; the reverse argument proves (16). Valid interval bounds $L_i \leq Y_i \leq U_i$ give projected companions with right-hand sides $\max\{0, \sum_{i \in A} (L_i - b_i)\}$ and $\max\{0, \sum_{i \in A} (b_i - U_i)\}$. Exact directed min-cuts separated these rows externally at the root. They strengthened many root relaxations but failed the frozen live gate, so they remain inactive research families.

4 Gurobi Fixed-Interval Implementation

The stable inner backend is a sparse tailored fixed-interval MILP solved by Gurobi’s native branch-and-cut. Gurobi manages presolve, LP relaxations, branching, node selection, incumbent processing, best bounds, and termination. Official deterministic settings are `Threads=1`, `Seed=0`, `Presolve=Auto`, `MIPGap=0`, and `MIPGapAbs=0`. Branching priorities are absent, `PreCrush` remains at its default, and no `MIPNODE` user-cut callback is registered.

The fixed-interval model is freshly constructed for each ledger leaf. Its canonical fingerprint binds variables, rows, domains, objective, input, solver settings, and executable identity. Bounds are accepted only when model scope, lifecycle, and fingerprint checks pass. Integer candidates are reconstructed and checked against original route, load, inventory, Gini, penalty, and objective definitions before they may update the incumbent.

Round 54’s inventory–route experiment was intentionally outside this stable path. It solved a fresh root LP, separated the most violated directed cut by an exact deterministic maximum-closure/min-cut construction with the depot fixed outside the station subset, added canonical nonduplicate rows, and rebuilt a fresh terminal MIP with integer types restored. Root closure had no hidden node, size, instance, or time dispatch. Invalid closure triggered F0 fallback. IR1 used the two mixed inequalities to full closure; IR2 added projected companions; IR3 was a predeclared one-pass option. Only IR1 entered the live comparison, and its negative result prevented integration into the stable preset.

Plain Gurobi is retained as a contextual benchmark. Its bound or certificate never enters the tailored interval ledger. Historical CPLEX-managed callback experiments are likewise contextual and are not the current implementation.

5 Certificate Ledger and Evidence Boundary

Each final Gini leaf records its interval, lower bound, closure basis, scope, model fingerprint, solver lifecycle, and coverage state. A global certificate requires a verified incumbent, complete improving-domain coverage, valid closure of every final leaf, no invalid interval or open node, and a monotone aggregate lower bound reaching the incumbent within tolerance.

Admissible leaf evidence consists of valid interval relaxations, exact fixed-interval infeasibility, scope-valid Gurobi best bounds, and empty-domain proofs. Same-run heuristic routes are upper

bounds only and must pass independent verification. Plain Gurobi, historical CPLEX rows, callback vectors, diagnostic models, archived incumbents, and wrapper checkpoints do not contribute to a K1-AM-SF certificate.

The Round 53 P-GRB benchmark originally omitted the pre-frozen expected-model- fingerprint binding used elsewhere. Round 54 preserved every original result, verified the original executable, froze 12 expected fingerprints before correction, and rebound the existing rows. Expected and actual fingerprints match for all 12; independent feasibility and objective checks pass. The strict count changes from 0/12 to 9/12, while the three native time-limit rows remain noncertificates. No Work, time, bound, gap, or GI value changed, and no F0 promotion conclusion changed.

Source-scope, command, fingerprint, certificate, evidence-hash, and package audits enforce this boundary. A finite-time bound is retained only with its honest noncertificate status. Current qualification supports V12 and bounded V20 use; V50 has startup/configuration evidence but lacks an opened Round 54 strengthening panel, so no universal scale claim is made.

6 Computational Protocol

Round 54 froze contracts, panels, promotion gates, source identity, and process caps before inspecting new performance. Ordinary processes were capped at 3,600 seconds; the predeclared 7,200-second V50 extension could open only after successful K1 integration. Every official fixed-interval comparator and candidate row used the same frozen Gurobi-enabled executable, one thread, seed zero, automatic presolve, zero MIP gaps, and complete process accounting.

The paper-preset identity audit used major, strong-control, V10 easy-negative, numerical-endpoint, V20, and V50 sentinels. All six have identical deterministic settings, controller actions, intervals, backend identity, objective, bound, and certificate class. Where model construction was reached, exported model files are byte-identical. The bounded V20/V50 runs remained in heuristic startup and produced no fixed-interval model file; this is reported as semantic command/controller evidence, not fabricated model evidence.

The inventory-route offline census covered 34 frozen states and both IR1 and IR2. Both variants were valid on every state, showed a strict violation in 33 structural roles, and improved the final root bound in 25 states. IR2 produced no strict final-closure gain over IR1, so the least expansive IR1 entered the live fixed-interval gate; the condition to open IR3 was absent.

Stage A compared F0-CLEAN and IR1 on D1-D14 at 300 seconds: 28 complete rows and 14 matched pairs. F0 certified 11/14 and IR1 9/14, with zero false certificates. IR1 gained one material hard-state improvement (D4), but lost the D1 and D13 certificates. Its shifted Work geometric-mean ratio was 1.089432; aggregate GI worsened from 0.498349 to 0.623423. The frozen Stage-B gate therefore failed. Fixed-interval Stage B, confirmation and long checks, full K1 integration, and the sealed V12/V20/V50 generalization runs were not opened. Empty schema-valid ledgers and opening decisions record this outcome.

The evidence supports K1-AM-SF as the stable mainline and classifies inventory-route strengthening as a bounded negative result. It does not support K1-AM-SF-IR or a universal paper-algorithm claim.

6.1 Round 54 fixed-interval gate

Method	Rows	Certificates	False certificates	Aggregate GI
F0-CLEAN	14	11	0	0.498349
IR1	14	9	0	0.623423

IR1/F0 shifted Work geometric mean is 1.089432. IR1 loses two baseline certificates and therefore fails the pre-frozen gate. Offline IR1 and IR2 each improve 25 of 34 root bounds, but that relaxation gain does not translate to nonworse terminal MIP performance in Stage A. Later gated stages have zero entered rows.

References

- [1] Corrado Gini. Variabilita e mutabilita. *Studi Economico-Giuridici della R. Universita di Cagliari*, 1912.